

Vision and Perception

Exam B

February 2023

Name, Surname, Student ID [please compile here]:

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The exam has to be carried out in 1 hour and 40 minutes.

In order to pass the exam you need to get 18 and a minimum of 8 points in each of the two parts. The exam consists in 6 exercises.

Write the answers for the two respective parts on two separate sheets.

1 Part 1

- **Exercise 1** Given an image Im and a filter f_1 , shows the intermediate passages and the the resulting image g after applying the convolution operator between Im and f_1 . Use the following coordinates (3,3),(1,1),(2,4) with zero padding. [4 points]

$$Im = \begin{bmatrix} 3 & 1 & 2 & 0 \\ 0 & 2 & 1 & 5 \\ 1 & 0 & 1 & 3 \\ 1 & 2 & 0 & 5 \end{bmatrix} \quad f_1 = \begin{bmatrix} -2 & 3 & 2 \\ 0 & -1 & -2 \\ 1 & -2 & 0 \end{bmatrix}$$

- **Exercise 2.a** List the main steps of the Harris corner detector.
Exercise 2.b Is the Harris corner detector robust with respect to uniform intensity changes in the image, i.e. uniform intensity shifts and intensity scaling? Justify your answer. Is the Harris corner detector robust with respect to rotation? Justify your answer. [6 points]
- **Exercise 3** Describe the optical flow algorithm. What is the aperture problem? What is the direction in the image along which optical flow cannot be reliably estimated? [6 points].

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2 Part 2

- **Exercise 4** Describe the camera calibration algorithm, reporting what is the input data, which quantities can be estimated with it and how it works (starting and ending equations will be sufficient) [6 points]
- **Exercise 5** Explain what are the epipoles e and e' in a two-view geometry, reporting also their characteristics, their main features, and the most important epipoles-related equations. [6 points]
- **Exercise 6** What is an hierarchical set of features in a Deep Learning model? Do we need to take any special action to build it when employing a deep convolutional neural network? [4 points]